Table 1.3 Sampling Distribution of Sample Mean

Sample mean: $\bar{x}$	4	6	8	10	12	14	16	Total
Probability: $P(\bar{X}=\bar{x})$	$\frac{1}{16}$	$\frac{2}{16}$	$\frac{3}{16}$	$\frac{4}{16}$	$\frac{3}{16}$	$\frac{2}{16}$	$\frac{1}{16}$	1

Note 4: The sample obtained under sampling *with replacement* from a finite population satisfies the conditions for a random sample as described earlier.

Note 5: If the sample values are selected under *without replacement scheme*, independence property of $X_1, X_2, \ldots X_n$ will be violated. Hence it will not be a random sample.

Note 6: When the sample size is greater than or equal to 30, in most of the text books, the sample is termed as a **large sample**. Also, the sample of size less than 30 is termed as **small sample**. However, in practice, there is no rigidity in this number *i.e.*, 30, and that depends on the nature of the population and the sample.

Note 7: The learners may recall from XI Standard Textbook that some of the probability distributions possess the additive property. For example, if $X_1, X_2, \ldots, X_n$ are *iid* $N(\mu, \sigma^2)$ random variables, then the probability distributions of $X_1 + X_2 + \ldots + X_n$ and $\bar{X}$ are respectively the $N(n\mu, n\sigma^2)$ and $N(\mu, \sigma^2/n)$. These two distributions, in statistical inference point of view, can be considered respectively as the sampling distributions of the sample total and sample mean of a random sample drawn from the $N(\mu, \sigma^2)$ distribution. The notation $N(\mu, \sigma^2)$ refers to the normal distribution having mean μ and variance σ^2.

1.3 STANDARD ERROR

The standard deviation of the sampling distribution of a statistic is defined as the **standard error** of the statistic, which is abbreviated as *SE*.

For example, the standard deviation of the sampling distribution of the sample mean, $\bar{x}$, is known as the standard error of the sample mean, or $SE(\bar{X})$.

If the random variables $X_1, X_2, \ldots, X_n$ are independent and have the same distribution with mean μ *and variance* σ^2, then variance of $\bar{X}$ becomes as

$$V(\bar{X}) = V\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\sum_{i=1}^{n} V(X_i) = \frac{1}{n^2}\sum_{i=1}^{n} \sigma^2 = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$$

Thus, $SE(\bar{X}) = \dfrac{\sigma}{\sqrt{n}}$.

Also, note that mean of $\bar{X} = E(\bar{X}) = E\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n}\sum_{i=1}^{n} E(X_i) = \frac{1}{n}\sum_{i=1}^{n} \mu = \frac{n\mu}{n} = \mu$

Example 1.2

Calculate the standard error of $\bar{X}$ for the sampling distribution obtained in *Example 1*.

Solution:

Here, the population is {4, 8, 12, 16}.

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